

Course: MAI511-1 - Advanced Database Technologies

LAB Exercise – V ACID Properties of Transaction

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Scenario

Student **201 (Vismitaa)** completes internship **301 (ML Intern)** and receives **95 marks with Grade A**.

1. Atomicity

Valid Transaction

```
UPDATE Internship_3NF SET internship_status='Completed'WHERE internship_id=301;
INSERT INTO Intern_Evaluation_NF VALUES(201,301,95,'A');
COMMIT;
```

Output:

```
1 row updated.

1 row inserted.

Commit complete.
```

Verification:

```
SELECT internship_id, internship_status FROM Internship_3NF WHERE internship_id=301;
```

Output for the verification:

```
INTERNSHIP_ID  INTERNSHIP_STAT
-----
301 Completed
```

```
SELECT *FROM Intern_Evaluation_NF WHERE student_id=201 AND internship_id=301;
```

Output:

STUDENT_ID	INTERNSHIP_ID	MARKS	G
201	301	95	A

Justification: Both operations were executed successfully and permanently stored after COMMIT.

Invalid Transaction

```
UPDATE Internship_3NF SET internship_status='Completed' WHERE internship_id=302;  
INSERT INTO Intern_Evaluation_NF VALUES(201,999,95,'A');  
ROLLBACK;
```

Output:

```
1 row updated.  
  
1 row inserted.  
  
Rollback complete.
```

Verification:

```
SELECT internship_status FROM Internship_3NF WHERE internship_id=302;
```

Output for verification:

```
INTERNSHIP_STAT  
-----  
Completed
```

Justification: Since Intern_Evaluation_NF currently has no foreign key constraint, it will actually allow 999.

2. CONSISTENCY

Valid Transaction

```
INSERT INTO Intern_Evaluation_NF VALUES(201,301,95,'A');  
COMMIT;
```

Output:

```
Commit complete.
```

Verification:

```
SELECT *FROM Intern_Evaluation_NF WHERE student_id=201 AND internship_id=301;
```

Output for the verification:

STUDENT_ID	INTERNSHIP_ID	MARKS	G
201	301	95	A

Invalid Transaction:

```
ALTER TABLE Intern_Evaluation_NF ADD CONSTRAINT chk_marks CHECK(marks BETWEEN 0 AND 100);  
INSERT INTO Intern_Evaluation_NF VALUES(201,301,150,'A');  
ROLLBACK;
```

Output:

```
Error starting at line : 134 in command -  
INSERT INTO Intern_Evaluation_NF VALUES(201,301,150,'A')  
Error report -  
ORA-02290: check constraint (SYS.CHK_MARKS) violated  
  
https://docs.oracle.com/error-help/db/ora-02290/  
  
More Details :  
https://docs.oracle.com/error-help/db/ora-02290/  
  
Rollback complete.
```

Verification:

```
SELECT *FROM Intern_Evaluation_NF WHERE marks=150;
```

Output for the Verification:

```
no rows selected
```

Justification: The CHECK constraint maintains database consistency.

3. ISOLATION

```

UPDATE Internship_3NF SET internship_status='Pending'WHERE internship_id=301;
UPDATE Internship_3NF SET internship_status='Completed'WHERE internship_id=301;
SELECT internship_status FROM Internship_3NF WHERE internship_id=301;
ROLLBACK;
SELECT Internship_status FROM Internship_3NF WHERE internship_id=301;

```

Output:

```

1 row updated.

1 row updated.

INTERNSHIP_STAT
-----
Completed

Rollback complete.

INTERNSHIP_STAT
-----
Pending

```

5. Durability

Valid Transaction

```

UPDATE Internship_3NF SET internship_status='Completed'WHERE internship_id=301;
COMMIT;
SELECT internship_status FROM Internship_3NF WHERE internship_id=301;

```

Output:

```

1 row updated.

Commit complete.

INTERNSHIP_STAT
-----
Completed

```

```
SELECT internship_status FROM Internship_3NF WHERE internship_id=301;
```

```
INTERNSHIP_STAT
```

```
-----
```

```
Completed
```

Justification: The internship status of Internship ID 301 was updated from Pending to Completed and the transaction was permanently stored using the COMMIT command. Even after closing and reopening the database session, the updated value remained as Completed. This demonstrates the Durability property of ACID, which ensures that once a transaction is committed, the changes are permanently stored in the database and are not lost even if the session is terminated or the system is restarted.